

**ILLINOIS INSTITUTE OF TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE**

**Database Organization
Course CS-425 02**

**Homework No.3
Case: Greendale Community College**

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Normalization to 3NF of Greendale Community College Database

To normalize the data given to the Third Normal Form (3NF), we need to eliminate redundancy and ensure that each attribute is dependent only on the primary key.

Assumptions

1. A student can enroll in multiple courses.
2. Each course is taught by one convenor.
3. The periods for each course are consistent across students.
4. On the Pre-Requisites and Periods columns, to reach normalization, we ensured that each attribute contains atomic (indivisible) values, and that the data does not contain any repeating groups or multi-valued attributes.
5. On the column of mark, it was assumed the value of '0' for symbols AB represents being absent from the exam, and DPR that denotes duly performed refusal – not permitted to write the examination in the course due to poor attendance or failing to meet the minimum requirement.

Entities and Attributes

Based on the database, the entities and their attributes can be defined as follows:

Student

- Each student has a unique identifier called StudentID.
- StudentName: The name of the student.
- PromotionCode: A code indicating the promotion status of the student (e.g., CON, REN, QUA).
- Qualify: A boolean value indicating whether the student qualifies (e.g., Y for yes, N for no).

Course

- Each course has a unique identifier called CourseID.
- CourseName: The name of the course.

Convenor

- Each convenor has a unique identifier called ConvenorID.
- ConvenorName: The name of the convenor.

Enrollment

- Each enrollment record is identified by a combination of StudentID, CourseID, and Year.
- Year: The year in which the student enrolled in the course.
- Mark: The mark the student received for the course.
- Symbol: A symbol representing the grade or status (e.g., 3, F, AB).

Pre-Requisite Type

- Each requisite type has a unique identifier called PreReqTypeID.
- TypeName: The name of the requisite type (e.g., All, One, Any, None).
- Symbol: A symbol representing the requisite type (e.g., &, One, OR, NULL).

Pre-Requisite

- Each Pre-requisite has a unique identifier called **PreReqID**.
- PreReqName: The name of the requisite course.

PeriodType

- Each period type has a unique identifier called PeriodTypeID.
- PeriodTypeName: The name of the period type (e.g., All, One, Any).
- PeriodTypeSymbol: A symbol representing the period type (e.g., AND, One, OR).

WeekDay

- Each weekday has a unique identifier called WeekDayID.
- WeekDay: The name of the weekday or the schedule type (e.g., Daily, Mon to Wed, every Tue & Fri, every Mon & Wed).

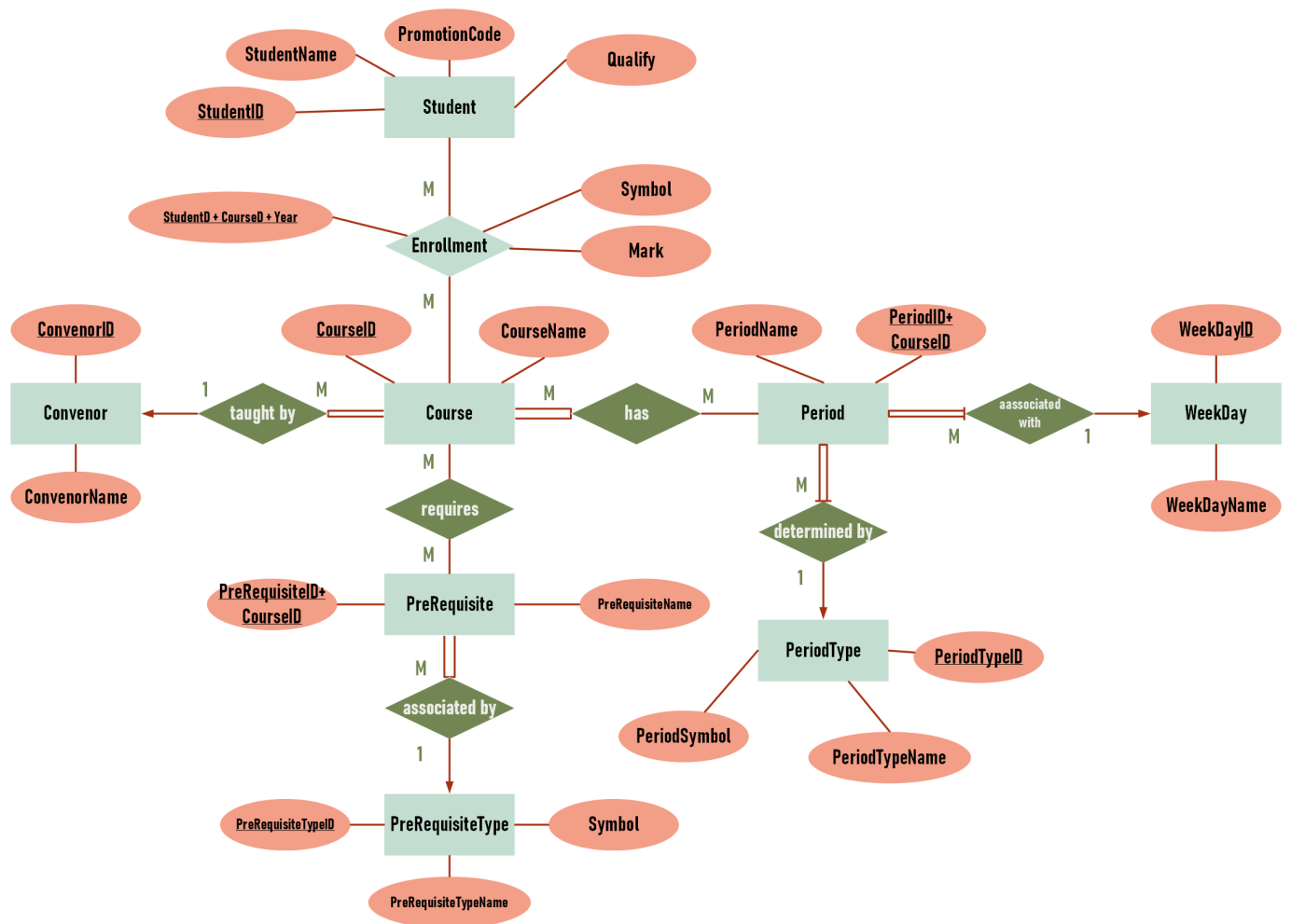
Course-PreRequisite

- Each course-prequisite relationship has a unique identifier called PreReqID.
- CourseID: The identifier of the course.
- PreReqTypeID: The identifier of the requisite type.

Course-Periods

- Each course-period relationship has a unique identifier called PeriodID.
- CourseID: The identifier of the course.
- PeriodTypeID: The identifier of the period type.
- WeekDayID: The identifier of the weekday or schedule type.

ER Diagram



Business Rules

Students and Enrollment:

- A student can enroll in zero or many (0..M) courses.
- A course can have zero or many (0..M) students enrolled in it.
- Business Rule: A student can enroll in (0..M) courses, and each course can have (0..M) students enrolled.

Courses and Convenors:

- A course must be taught by one and only one (1..1) convenor.
- A convenor can teach zero or many (0..M) courses.
- Business Rule: Each course must be taught by exactly one convenor, but a convenor can teach (0..M) courses.

Courses and PreRequisites:

- A course can have zero or many (0..M) prerequisites.
- A prerequisite can be a requirement for zero or many (0..M) courses.
- Business Rule: A course can have (0..M) prerequisites, and each prerequisite can be required by (0..M) courses.

PreRequisites and PreRequisiteType:

- A prerequisite must be associated with one and only one (1..1) prerequisite type.
- A prerequisite type can be used by zero or many (0..M) prerequisites.
- Business Rule: Each prerequisite must have exactly one type, but a type can be assigned to (0..M) prerequisites.

Courses and Periods:

- A course can have one or many (1..M) periods.
- A period can be assigned to zero or many (0..M) courses.
- Business Rule: A course must have at least one period (1..M), and a period can be shared by (0..M) courses.

Periods and PeriodType:

- A period must be associated with one and only one (1..1) period type.
- A period type can be used by zero or many (0..M) periods.
- Business Rule: Each period must have exactly one type, but a type can be assigned to (0..M) periods.

Periods and WeekDay:

- A period must be scheduled with one and only one (1..1) weekday.
- A weekday can be used by zero or many (0..M) periods.
- Business Rule: Each period must occur on exactly one weekday, but a weekday can have (0..M) periods.

Logical Design - Relation Schemas

Student(StudentID (PK), StudentName, Qualify, PromotionCode)

Convenor(ConvenorID (PK), ConvenorName)

Course(CourseID (PK), CourseName, ConvenorID (FK))

Enrollment(StudentID (FK), CourseID (FK), Year, Mark, Symbol, PRIMARY KEY (StudentID, CourseID, Year))

PreRequisiteType(PreReqTypeID (PK), TypeName, Symbol)

PreRequisite(PreReqID (PK), PreReqName)

CoursePreRequisite(PreReqID (FK), CourseID (FK), PreReqTypeID (FK),
PRIMARY KEY (PreReqID, CourseID))

PeriodType(PeriodTypeID (PK), PeriodTypeName, PeriodTypeSymbol)

WeekDay(WeekDayID (PK), WeekDay)

Period(PeriodID (PK), PeriodName)

CoursePeriod(PeriodID (FK), CourseID (FK), PeriodTypeID (FK), WeekDayID (FK),

PRIMARY KEY (PeriodID, CourseID))

Functional Dependencies

- **Student (StudentID, StudentName, PromotionCode, Qualify)**
 - Full Functional Dependencies: StudentID → Name, PromotionCode, Qualify
 - There are no partial or transitive dependencies as StudentID is the primary key and all attributes depend fully on StudentID.
- **Course (CourseID, CourseName, ConvenorID, PreRequisites, Periods)**
 - Full Functional Dependencies: CourseID → CourseName, ConvenorID, PreRequisites, Periods
 - There are no partial dependencies as CourseID is the primary key and all attributes depend fully on CourseID.
- **Convenor (ConvenorID, ConvenorName)**
 - Full Functional Dependencies: ConvenorID → ConvenorName
 - There are no partial or transitive dependencies as ConvenorID is the primary key and ConvenorName depends fully on ConvenorID.
- **Enrollment (StudentID, CourseID, Year, Mark, Symbol)**
 - Full Functional Dependencies: (StudentID, CourseID, Year) → Mark, Symbol
 - Partial Dependencies: StudentID → Name, PromotionCode, Qualify (if we assume these are included in the Enrollment schema)
 - There are no transitive dependencies within the Enrollment schema.

Keys

- **Primary Keys (PK), Candidate Keys and Surrogate Keys:** StudentID, ConvenorID, CourseID, PreReqTypeID, PreReqID, PeriodTypeID, WeekDayID, PeriodID. In this context the candidate keys and surrogate keys are the same as the primary keys.
- **Composite PKs** in CoursePreRequisite, CoursePeriod, and Enrollment tables.
- **Composite Keys:** Composite PKs in CoursePreRequisite, CoursePeriod, and Enrollment tables.
- **Foreign Keys (FK):** ConvenorID in Course Table, PreReqID and CourseID in CoursePreRequisite Table, PeriodID and CourseID in CoursePeriod Table, StudentID and CourseID in Enrollment Table.

Normal Forms

1st Normal Form (1NF):

- The relation schemas are in 1NF because all attributes are atomic, and there are no repeating groups.

2nd Normal Form (2NF)

- The relation schemas are in 2NF because they are in 1NF and all non-key attributes are fully functional dependent on the primary key.

3rd Normal Form (3NF):

- The relation schemas are in 3NF because:
 - They are in 2NF (all non-key attributes are fully functional dependent on the primary key).
 - There are no transitive dependencies, meaning non-key attributes are not dependent on other non-key attributes.

Design Decisions and Limitations

- The Convenor table is separated to avoid redundancy in storing convenor names.
- The Enrollment table handles the many-to-many relationship between students and courses.
- This design assumes that each course is uniquely identified by CourseID and that periods for a course are the same for all students enrolled in it.

Data in the New Schema Excel Document